

JEE MAIN REVISION MASTER: ROTATIONAL MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Torque, Angular Momentum & Rotational Kinetic Energy	7 PYQs	Rotational Dynamics
Moderate	Moment of Inertia & Theorems of Parallel / Perpendicular Axes	6 PYQs	Rotational Inertia

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Torque, Angular Momentum & Rotational Kinetic Energy	$\tau = I\alpha$, $L = I\omega$, $K_{\text{rot}} = \frac{1}{2}I\omega^2$, $I_1\omega_1 = I_2\omega_2$
Moment of Inertia & Theorems of Parallel / Perpendicular Axes	$I_{\text{disc}} = \frac{1}{2}MR^2$, $I_{\text{solid sphere}} = \frac{2}{5}MR^2$, $I_{\text{tangent}} = I_{\text{cm}} + Md^2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Torque, Angular Momentum & Rotational Kinetic Energy (Moderate Yield — 7 PYQs)

Core Principle: Rotational dynamics: Torque $\tau = I\alpha = (dL / dt)$. Angular momentum $L = I\omega = \text{vec}(r) \times \text{vec}(p)$. If external torque is zero ($\tau_{\text{ext}} = 0$), angular momentum is conserved ($I_1\omega_1 = I_2\omega_2$). Rotational kinetic energy $K_{\text{rot}} = \frac{1}{2}I\omega^2$.

■■ **Caution / JEE Pitfall:** When a ballet dancer pulls in her arms, I decreases, ω increases, but rotational kinetic energy **INCREASES** due to internal muscular work done!

■ **Speed Trick:** Convert rpm to rad/s: $\omega = (2\pi N / 60)$. A common 1-step calculation in KCET rotational problems.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The angular momentum of a moving body remains constant, if..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1=4\text{kgm}^2, r_1=20\text{cm}$, $I_2=20\text{kgm}^2$ and $r_2=30\text{cm}$..." [Ans: N/A]

3.2 Moment of Inertia & Theorems of Parallel / Perpendicular Axes (Moderate Yield — 6 PYQs)

Core Principle: Moment of inertia $I = \sum m_i r_i^2 = M k^2$. Parallel axis theorem: $I = I_{cm} + M d^2$. Perpendicular axis theorem for laminar bodies: $I_z = I_x + I_y$. Standard values: Ring about axis $M R^2$; Disc $(1/2) M R^2$; Solid Sphere $(2/5) M R^2$; Thin Rod $(1/12) M L^2$.

■ **Caution / JEE Pitfall:** Perpendicular axis theorem applies ONLY to flat planar (2D) lamina, NEVER to 3D objects like solid spheres or cylinders!

■ **Speed Trick:** For rolling bodies down an incline, acceleration is $a = (g \sin \theta / 1 + k^2/r^2)$. Smaller k^2/r^2 reaches bottom first: Solid Sphere (0.4) > Disc (0.5) > Ring (1.0).

Representative Source PYQs:

- **[2024 Shift] Q#4 (MCQ):** "A ceiling fan is rotating around a fixed axle as shown. The direction of angular velocity is along" [Ans: N/A]
- **[2023 Shift] Q#2 (MCQ):** "The moment of inertia of a rigid body about an axis..." [Ans: N/A]
- **[2023 Shift] Q#3 (MCQ):** "Seven identical discs are arranged in a planar pattern, so as to touch each other as shown in the figure. Each disc has mass m radius R ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?

- (A) 6 h
- (B) 18 h
- (C) 24 h
- (D) 30 h

Correct Answer: None

Solution Analysis: When the earth contracts suddenly, no external torque is acting on it. So its angular momentum remains conserved. Angular momentum is $L = I\omega$. For a spherical body, $I \propto R^2$. If the radius becomes half, $R' = (R/2)$. So the new moment of inertia becomes $I' = I((1/2))^2 = (I/4)$. Using conservation of angular momentum, $I\omega = I'\omega'$. $I\omega = (I/4)\omega'$. $\omega' = 4\omega$. So the earth will rotate 4 times faster. Time period is inversely proportional to angular velocity, so $T' = (T/4)$. Present duration of a day is $T = 24$ h. Hence, $T' = 6$ h.

Benchmark Problem 2: KCET 2023 (Q#2)

The moment of inertia of a rigid body about an axis

- (A) does not depend on its mass.
- (B) does not depend on its shape.
- (C) depends on the position of axis of rotation.
- (D) does not depend on its size.

Correct Answer: None

Solution Analysis: Moment of inertia of a body depends on the mass of the body, distribution of mass in the body, position of axis of rotation of the body and on the distance from the axis of rotation.